EASY · ADVANCED MATH · ANSWER KEY

Advanced Math

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ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1

D

D. $4t^2s16s^2 - 14t$

Choice D is correct. For the expression $64t^2s^3 - 56t^3s$, 4 is a common factor of 64 and 56, t^2 is a common factor of t^2 and t^3, and s is a common factor of s^3 and s. It follows that $4t^2s$ is a common factor of each term in the given expression. Factoring out $4t^2s$ from each term in the expression $64t^2s^3 - 56t^3s$ yields $4t^2s(16s^2 - 14t)$. Therefore, $4t^2s(16s^2 - 14t)$ is equivalent to $64t^2s^3 - 56t^3s$.

Choice A is incorrect. This expression is equivalent to $64ts^3 - 56t^2s^2$, not $64t^2s^3 - 56t^3s$.

Choice B is incorrect. This expression is equivalent to $64t^3s^3 - 56t^2s$, not $64t^2s^3 - 56t^3s$.

Choice C is incorrect. This expression is equivalent to $64t^3s^2 - 56t^2s^2$, not $64t^2s^3 - 56t^3s$.

Q2

-3

The correct answer is -3. The y-intercept of the graph of a function in the xy-plane is the point where the graph crosses the y-axis. The graph of the polynomial function shown crosses the y-axis at the point $0, -3$. It's given that the y-intercept of the graph is $0, y$. Thus, the value of y is -3.

Q3

A

A. 432

Choice A is correct. The first equation in the given system of equations is $x = 3$. Substituting 3 for x in the second equation in the given system of equations yields $y = 15 - 3^2$, or $y = 144$. Substituting 3 for x and 144 for y in the expression xy yields $(3)(144)$, or 432. Therefore, the value of xy is 432.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q4

C

C. $34x + 9$

Choice C is correct. Each term in the given expression, $12x + 27$, has a common factor of 3. Therefore, the expression can be rewritten as $34x + 39$, or $34x + 9$. Thus, the expression $34x + 9$ is equivalent to the given expression.

Choice A is incorrect. This expression is equivalent to $108x + 12$, not $12x + 27$.

Choice B is incorrect. This expression is equivalent to $324x + 27$, not $12x + 27$.

Choice D is incorrect. This expression is equivalent to $27x + 72$, not $12x + 27$.

Q5

B

B. $0, 2$

Choice B is correct. The y-intercept of a graph in the xy-plane is the point at which the graph crosses the y-axis. The graph shown crosses the y-axis at the point $0, 2$. Therefore, the y-intercept of the graph shown is $0, 2$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q6

A

A. $c = 25p + k + 7$

Choice A is correct. Adding 7 to each side of the given equation yields $c = 25p + k + 7$.

Choice B is incorrect. This equation is equivalent to $c + 7 = 25p + k$, not $c - 7 = 25p + k$.

Choice C is incorrect. This equation is equivalent to $\frac{c}{7} = 25p + k$, not $c - 7 = 25p + k$.

Choice D is incorrect. This equation is equivalent to $7c = 25p + k$, not $c - 7 = 25p + k$.

Q7

A

A. $56y^2z^2$

Choice A is correct. The given expression can be rewritten as $8 \cdot 7y \cdot yz \cdot z$, which is equivalent to $56y^2z^2$, or $56y^2z^2$.

Choice B is incorrect. This expression is equivalent to $8yzy7$.

Choice C is incorrect. This expression is equivalent to $8zy7$.

Choice D is incorrect and may result from conceptual or calculation errors.

Q8

C

C. 5

Choice C is correct. It's given that the function g is defined by $gx = \sqrt{8x+1}$. Substituting 3 for x in the given function yields $g3 = \sqrt{83+1}$, which is equivalent to $g3 = \sqrt{25}$, or $g3 = 5$. Therefore, the value of $g3$ is 5.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of $83 + 1$, not $\sqrt{83+1}$.

Q9

D

D. 72

Choice D is correct. Since the graphs of the equations in the given system intersect at the point x, y, the point x, y represents a solution to the given system of equations. The first equation of the given system of equations states that $x = 8$. Substituting 8 for x in the second equation of the given system of equations yields $y = 8^2 + 8$, or $y = 72$. Therefore, the value of y is 72.

Choice A is incorrect. This is the value of x, not y.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q10 C

C. $x23x^2 + 2x + 9$

Choice C is correct. Since x is a common factor of each term in the given expression, the given expression can be rewritten as $x23x^2 + 2x + 9$.

Choice A is incorrect. This expression is equivalent to $23x^3 + 46x^2 + 207x$.

Choice B is incorrect. This expression is equivalent to $207x^4 + 18x^3 + 9x$.

Choice D is incorrect. This expression is equivalent to $34x^3 + 34x^2 + 34x$.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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